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CAPITAL UNIVERSITY OF ECONOMICS AND BUSINESS

TimeXer: Empowering Transformers for Time Series Forecasting with Exogenous Variables

TimeXer：通过外生变量来加强Transformer进行时间序列预测

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Abstract

Recent studies have demonstrated remarkable performance in time series forecasting. However, due to the partially-observed nature of real-world applications, solely focusing on the target of interest, so-called *endogenous variables*, is usually insufficient to guarantee accurate forecasting. Notably, a system is often recorded into multiple variables, where the *exogenous series* can provide valuable external information for endogenous variables. Thus, unlike prior well-established multivariate or univariate forecasting that either treats all the variables equally or overlooks exogenous information, this paper focuses on a practical setting, which is time series forecasting with exogenous variables. We propose a novel framework, **TimeXer**, to utilize external information to enhance the forecasting of endogenous variables. With a deftly designed embedding layer, TimeXer empowers the canonical Transformer architecture with the ability to reconcile endogenous and exogenous information, where patch-wise self-attention and variate-wise cross-attention are employed. Moreover, a global endogenous variate token is adopted to effectively bridge the exogenous series into endogenous temporal patches. Experimentally, TimeXer significantly improves time series forecasting with exogenous variables and achieves consistent state-of-the-art performance in twelve real-world forecasting benchmarks.

TimeXer: Empowering Transformers for Time Series Forecasting with Exogenous Variables

- 问题描述

Endogenous v.s. Exgenous

内生变量endogenous variables:仅仅关注自身感兴趣的变量
外生变量exogenous variables：为内生变量提供有用的外部信息
目的：把多变量预测转化成单变量预测，通过外生变量对内生变量提供更多辅助信息。通过模型框架协调内生和外生信息

- 作者的工作

——提出模型框架：
Time Series Transform**er** with e**X**ogenous variables (**TimeXer**)
在原有的transformer架构基础上，引入patch-wise self attention 以及variate-wise cross-attention

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01 Introduction

以后续实验数据集中长期预测的数据集为例，解释外生变量和内生变量的定义

（1）ECL：来自321个客户的每小时用电量数据
内生变量：最后一个客户的用电量
外生变量：其他客户的用电量
（2）Weather：气象站每10分钟所采集的21个气象因子
内生变量：Wet Bulb因子
外生变量：其他指标（如二氧化碳浓度等
（3）ETT：变压器油温数据（包含四个子集，由每小时记录次数划分
内生变量：油温
外生变量：6个功率负荷特征
（4）Traffic：某高速公路的862个传感器测量的每小时道路占用率
内生变量：最后一个传感器的测量值
外生变量：其他传感器的测量值

例如，电价高度依赖于市场的供求关系，仅根据历史数据几乎不可能预测未来的价格。

01 Introduction

如果不考虑外生变量的影响的问题：

（1）将外生变量与内生变量**等同看待**不仅会导致显著的时间和记忆复杂性，而且会涉及内生序列与外部信息之间不必要的相互作用。

（2）外部因素对内生序列的影响可能是连续的和时滞的

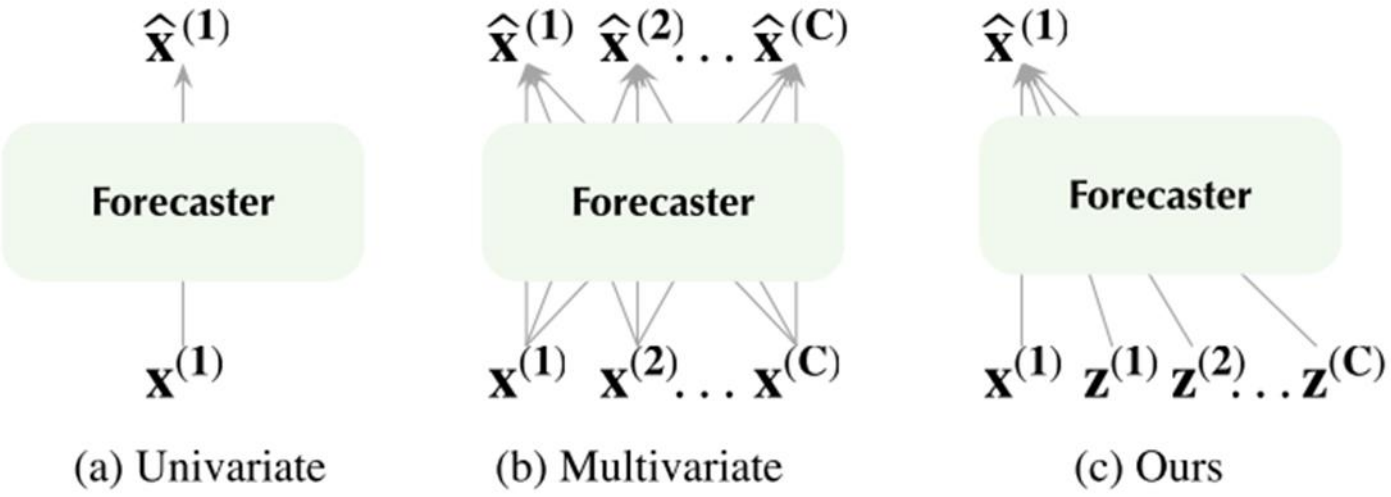


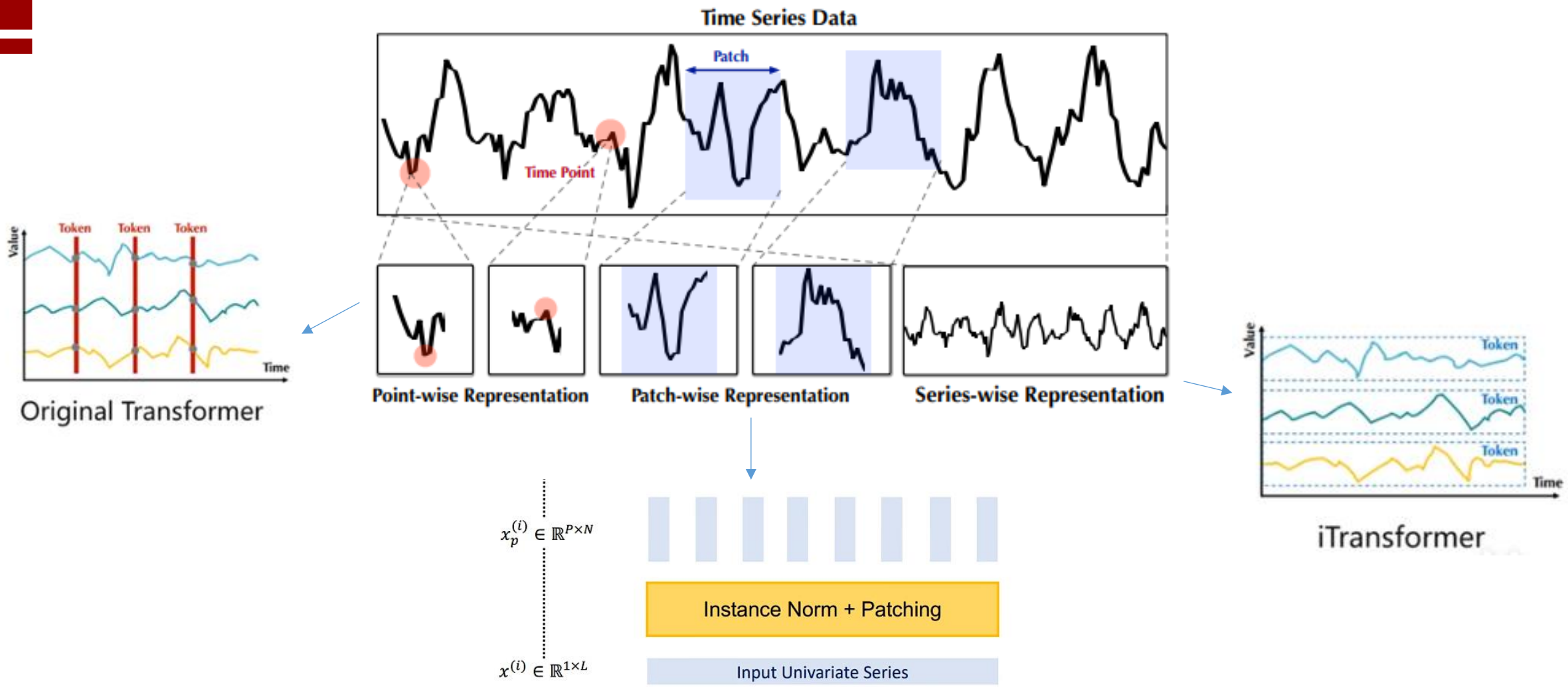
Figure 1. Comparison between different forecasting paradigms. The inputs of forecasting with exogenous variables include multiple external variates $\{\mathbf{z}^{(1)}, \cdots, \mathbf{z}^{(C)}\}$, which serve as auxiliary information and are not required for forecasting.

01 创新点

先前模型的缺陷：
以PatchTST为代表，它只能捕捉**时间依赖性**，而不能捕捉**多元相关性**。
相反地，iTransformer把一个序列看成一个token，通过在变量token上使用注意机制来捕获变量之间的相互关系，但是不能捕获不同子序列之间的时间变化。

TimeXer:
(1) 在包含外生变量影响的基础上，不去修改transformer的架构
(2) **Self-attention：**对patch-level的内生时序token提取时间相关性，捕获时间依赖性
(3) **CrossAttention:**对变量token，捕捉多变量之间的相关性，研究内生与外生变量之间的相关性，解决内生与外生变量之间的关系

02 Related Work



通过利用时间序列数据的序列级表示，TimeXer可以通过交叉注意力捕获多个外生变量和内生变量之间的相关性。

逐点表示在揭示局部语义信息方面存在不足

03 Method

Problem Definition Given an endogenous time series 1. $\mathbf{x}_{1:L} = \{x_1, x_2, ..., x_L\} \in \mathbb{R}^{L\times 1}$ and corresponding exogenous series 2. $\mathbf{z}_{1:L'} = \{\mathbf{z}_{1:L'}^{(1)}, \mathbf{z}_{1:L'}^{(2)}, ..., \mathbf{z}_{1:L'}^{(C)}\} \in \mathbb{R}^{L\times C}$, where L is the length of the look-back window of the endogenous variable, x_i denotes the value at the i-th time points, L' is the length of the look-back window of exogenous variables, C is the number of exogenous variables, and $\mathbf{z}_{1:L'}^{(i)}$ represents the i-th exogenous variable. Note that the length of endogenous series L and exogenous series L' can be different. The goal of forecasting model f is to predict the future S time steps $\widehat{\mathbf{x}} = \{x_{L+1}, x_{L+2}, ..., x_{L+S}\}$ based on both historical observations $\mathbf{x}_{1:L}$ and corresponding exogenous series $\mathbf{z}_{1:L'}$, which can be formulated as follows:

3. $$\widehat{\mathbf{x}} = f\left(\mathbf{x}_{1:L}, \mathbf{z}_{1:L'}\right). \tag{1}$$

其中L和L'可以不相等

只预测内生时间序列，外生变量是附加因素

03 Method

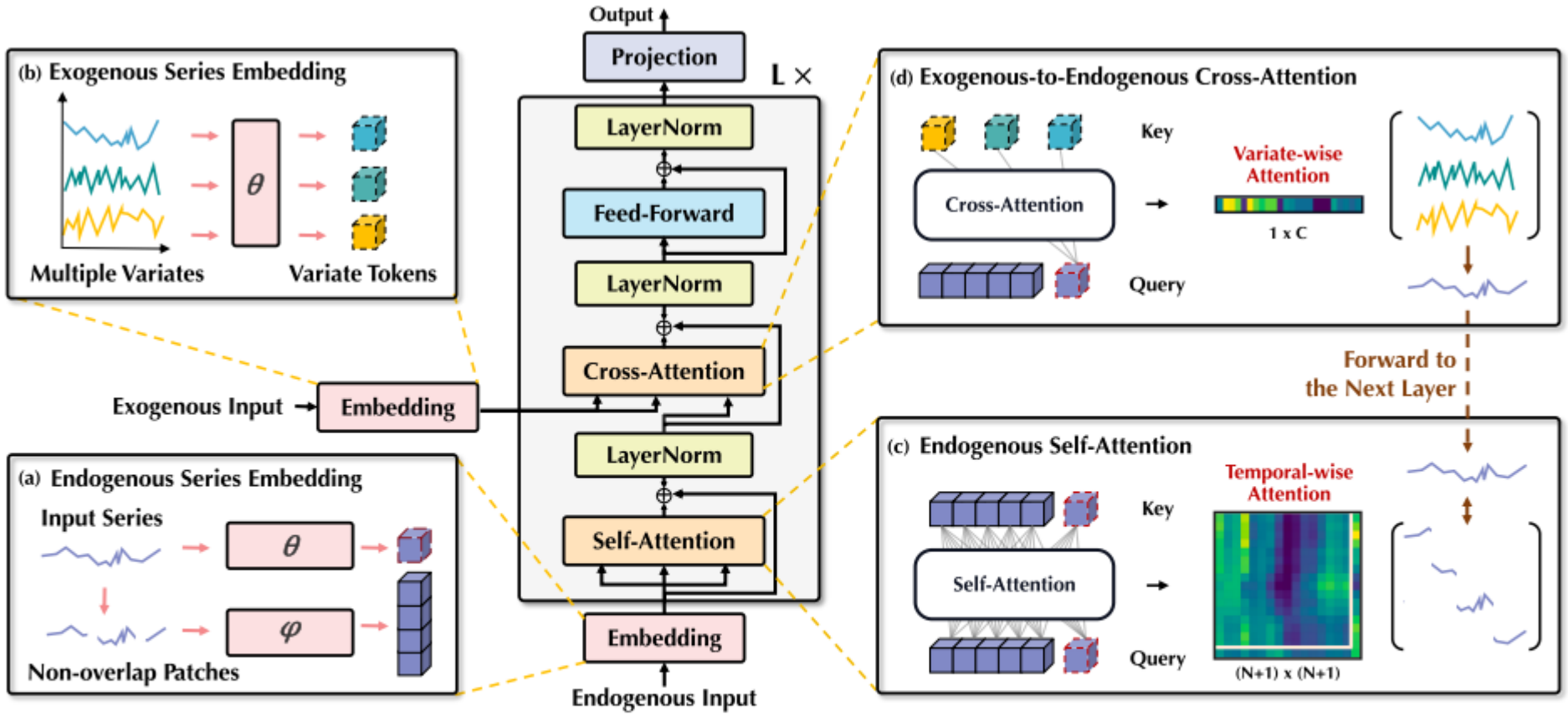
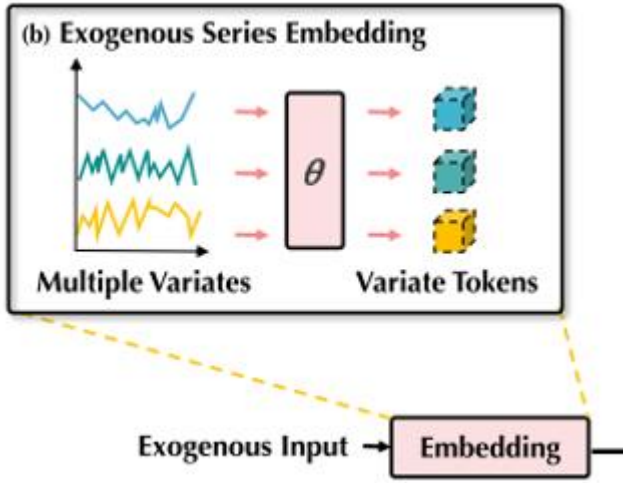
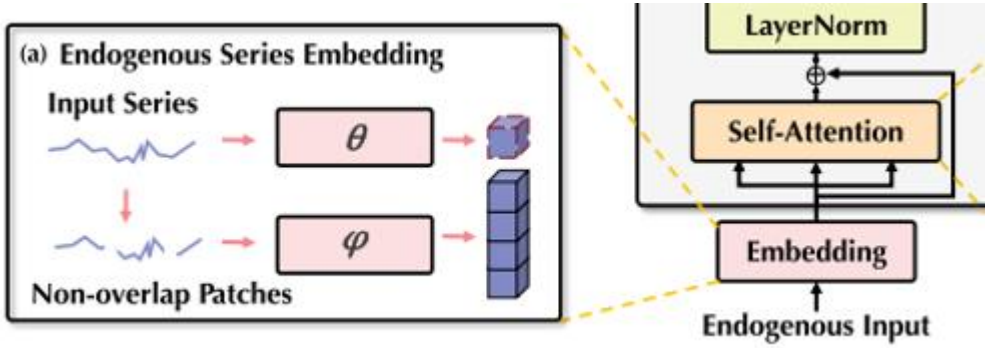


Figure 3. The overall structure of TimeXer. TimeXer empowers the canonical Transformer to the processing of exogenous variables. (a) Both patch embedding and variate embedding are utilized for the endogenous variables to obtain multiple temporal tokens and a variate token respectively. (b) Each exogenous variable is embedded as a variate token through variate embedding. (c) Self-attention is applied to the endogenous temporal tokens to capture patch-wise dependencies. (d) Cross-attention is adopted to model the series-level dependencies over endogenous and exogenous variables.

03 Variate Embedding





■ 变量的嵌入方式（基于itransformer

- 对于外生变量：每一个Series记作一个token
- 全局标记：利用内生序列和外生序列的序列全局标记来模拟不同变量之间的相互作用（代表整个序列的信息
- 经过线性投影将每个序列嵌入到一个变量token中，以实现变量嵌入

$$\mathbf{V}_{en} = \mathrm{EnVariateEmbed}\left(\mathbf{x}\right), \rightarrow \mathbb{R}^{L} \rightarrow \mathbb{R}^{D}$$

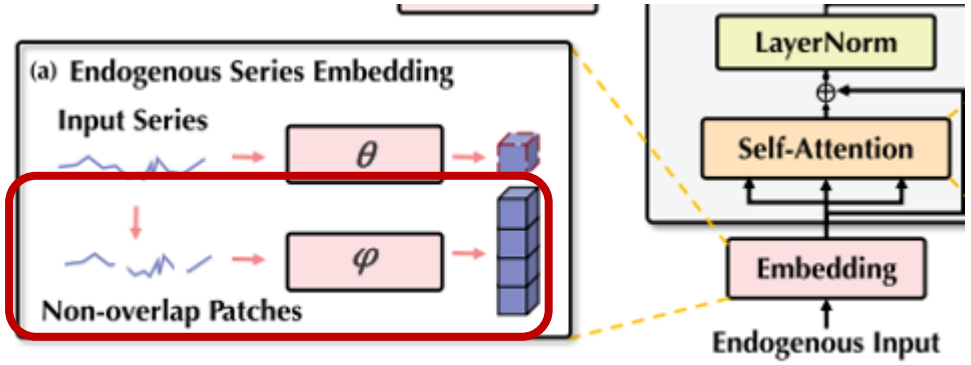
$$\mathbf{V}_{ex,i} = \mathrm{ExVariateEmbed}\left(\mathbf{z}^{(i)}\right) \; i \in \{1, \cdots, C\}. \quad (2) \rightarrow \mathbb{R}^{L'} \rightarrow \mathbb{R}^{D}$$

stantiated as trainable linear projectors, where L and L' are the look-back length of endogenous and exogenous series respectively. In this way, the embedded vector contains information of the whole series. $\mathbf{V}_{ex} = \{\mathbf{V}_{ex,i}\}_{i=1}^{C}$ denotes the set of representations for multiple exogenous series.

03 Patch Embedding

- **Input 嵌入（基于PatchTST**
Patch-token+Non-overlap

- **线性层+PE（Positional Embedding） P ⇒ D**
先分patch，然后每个patch做嵌入，得到N个patch



将内生序列分割为长度为 P 的补丁，并将每个补丁映射为一个补丁标记：

$$\{\mathbf{s}_1, \mathbf{s}_2, ..., \mathbf{s}_N\} = \text{Patchify}(\mathbf{x}),$$
$$\mathbf{P}_{en} = \text{PatchEmbed}(\mathbf{s}_1, \mathbf{s}_2, ..., \mathbf{s}_N). \quad (3)$$

$$\mathbf{P}_{en} \in \mathbb{R}^{N \times D} \qquad N = \lfloor \frac{L}{P} \rfloor$$

Denote the patch length as P, $N = \lfloor \frac{L}{P} \rfloor$ is the number of patches of the endogenous series, and $\mathbf{s}_i$ denotes the i-th patch in the patch sequence. The PatchEmbed() map each patch of length P into D dimensions with a trainable linear projection and position embedding. In this way, the

P:patch的长度，N：总Patch的个数

03 Patch-wise Self-Attention

variables contains multiple patch-wise temporal tokens $\widehat{\mathbf{P}}_{en}$ and a series-wise variate token $\widehat{\mathbf{V}}_{en}$, where the variate token serves as a global token of the whole series, incorporating information from other temporal tokens. This provides a comprehensive view of the temporal dependencies within the endogenous variable, as well as better interaction with the series-wise variate tokens of the exogenous variables. The process can be formalized as:

$$\widehat{\mathbf{P}}_{en}^{l}, \widehat{\mathbf{V}}_{en}^{l} = \mathrm{LN}\left([\mathbf{P}_{en}^{l}, \mathbf{V}_{en}^{l}] + \text{Self-Attn}\left([\mathbf{P}_{en}^{l}, \mathbf{V}_{en}^{l}]\right) \right), \tag{4}$$

where $l \in \{0, \cdots, M-1\}$ denotes the l-th TimeXer block. $\mathbf{P}_{en}^{l}$ and $\mathbf{V}_{en}^{l}$ are the patch-wise and variate-wise token of the endogenous variable respectively, where $\mathbf{P}_{en}^{0}, \mathbf{V}_{en}^{0}$ are the embedded $\mathbf{P}_{en}, \mathbf{V}_{en}$. Here, $[\cdot,\cdot]$ denotes the concatenation of the patch-wise and series-wise tokens of the endogenous variable along the sequence dimension. Self-Attn$(\cdot)$ denotes the Multi-head Self-Attention over the endogenous tokens. LN$(\cdot)$ denotes Layer Normalization.

➢ 捕捉内生变量内在的时间依赖性

将内生变量的patch token和series token（全局变量）进行concat

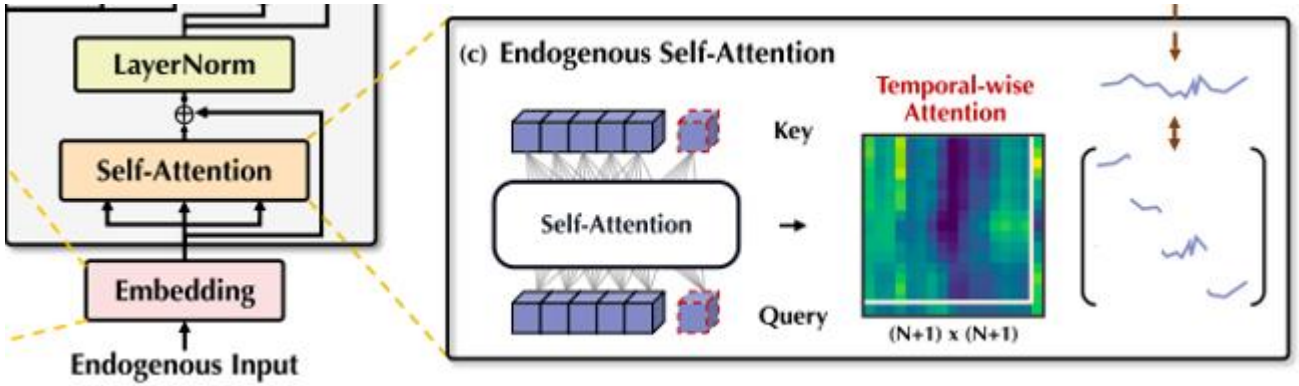
做注意力机制

注意：矩阵维度为(N+1)×(N+1) → N个patch+一个series

➢ 实现

Concatenation: Patch-wise token Series-wise token

Self Attention+残差关系+LN归一化层



03 Variate-wise Cross-Attention

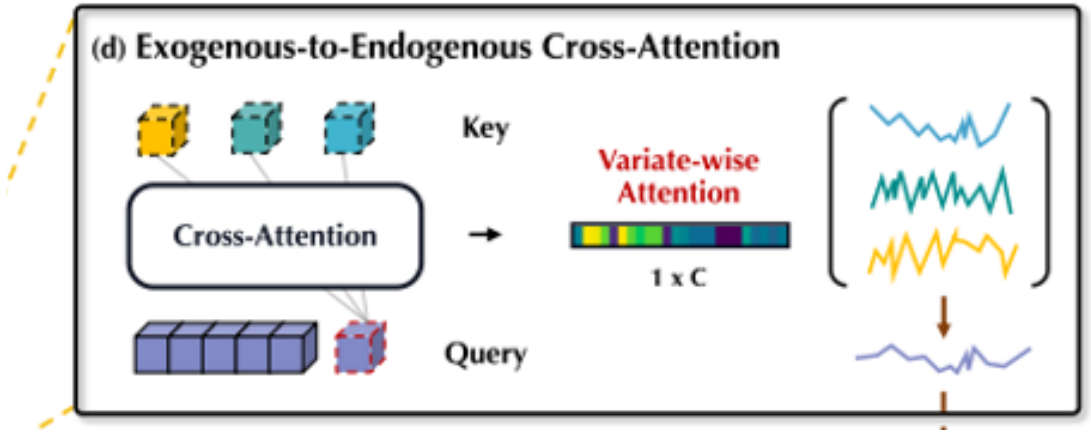
■ 去探究内生变量的内在时间依赖性（不预测外生变量）

- Query:内生变量的series-wise的自注意力机制的结果，其中内生变量还是patch+series
- Key & Value：外生变量的token

从而建立两类变量之间的联系

■ Cross Attention+残差关系+LN归一化层实现：

$$\mathbf{V}_{en}^{l+1} = \mathrm{LN}\left(\widehat{\mathbf{V}}_{en}^{l} + \text{Cross-Attn}\left(\widehat{\mathbf{V}}_{en}^{l}, \mathbf{V}_{ex}, \mathbf{V}_{ex}\right)\right).$$



交叉注意力得到的输出，将成为下一个self-attention层的输入，内生变量token学习到的多元相关性可以进一步扩散，实现外生变量token和内生时间token之间的信息传递。

03 Forecasting Loss

Forecasting Loss As aforementioned in the problem definition, since the exogenous variables do not need to be predicted by the model, we apply a linear projection to the decoder output to yield the prediction of endogenous variables. Finally, we opt to use the L2 loss to measure the discrepancy between the prediction and the ground truth:

$$\begin{aligned} \widehat{\mathbf{x}} &= \text{Projection}(\mathbf{V}_{en}^{L}), \\ \text{Loss} &= \sum_{i=1}^{S} \|\mathbf{x}_i - \widehat{\mathbf{x}}_i\|_2^2. \end{aligned} \tag{6}$$

where L is the number of TimeXer block, $\widehat{\mathbf{x}} \in \mathbb{R}^{S \times 1}$ is the prediction of future S time steps of the endogenous variable.

04 Datasets

实验涵盖了多个真实世界的时间序列数据集，包括短期和长期预测任务。这些数据集涉及电力市场、气候、交通等多个领域，以全面评估TimeXer在不同场景下的表现。

Table 5. Dataset descriptions. *Ex.* and *En.* are abbreviations for the Exogenous variable and Endogenous variable, respectively. The dataset size is organized in (Train, Validation, Test)

	Dataset	#Num	Ex. Descriptions	En. Descriptions	Sampling Frequency	Dataset Size
长期	Electricity	320	Electricity Consumption	Electricity Consumption	1 Hour	(18317, 2633, 5261)
	Weather	20	Climate Feature	CO2-Concentration	10 Minutes	(36792, 5271, 10540)
	ETTh	6	Power Load Feature	Oil Temperature	1 Hour	(8545, 2881, 2881)
	ETTm	6	Power Load Feature	Oil Temperature	15 Minutes	(34465, 11521, 11521)
	Traffic	861	Road Occupancy Rates	Road Occupancy Rates	1 Hour	(12185, 1757, 3509)
短期	NP	2	Grid Load, Wind Power	Nord Pool Electricity Price	1 Hour	(36500, 5219, 10460)
	PJM	2	System Load, SyZonal COMED load	Pennsylvania-New Jersey-Maryland Electricity Price	1 Hour	(36500, 5219, 10460)
	BE	2	Generation, System Load	Belgium's Electricity Price	1 Hour	(36500, 5219, 10460)
	FR	2	Generation, System Load	France's Electricity Price	1 Hour	(36500, 5219, 10460)
	DE	2	Wind power, Ampirion zonal load	German's Electricity Price	1 Hour	(36500, 5219, 10460)

对于短期预测任务，是五个主要市场的电价短期预测数据集，每个数据集包含电价作为一个内生变量和两个在实践中有影响的外生变量。
至于长期预测，我们采用了七个成熟的公共长期多元预测基准进行预测与外生变量。

04 Short-term forecasting task

模型细节：补丁长度为24,并遵循短期电价预测中的标准协议,其中输入长度和预测长度被设置为168和24。较低的MSE或MAE表示更准确的预测结果。

Table 2. Full results of the short-term forecasting task on EPF dataset. We follow the standard protocol in short-term electricity price forecasting, where the input length and predict length are set to 168 and 24 respectively for all baselines. Avg means the average results from all five datasets. The character "." in the Transformers indicates the name of *former.

Model	TimeXer (Ours)		iTrans. (2023)		RLinear (2023)		PatchTST (2022)		Cross. (2022)		TiDE (2023)		TimesNet (2023a)		DLinear (2023)		SCINet (2022a)		Stationary (2022b)		Auto. (2021)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
NP	**0.238**	**0.268**	0.265	0.300	0.335	0.340	0.267	0.284	0.245	0.289	0.335	0.340	0.250	0.289	0.309	0.321	0.373	0.368	0.294	0.308	0.402	0.398
PJM	**0.088**	**0.188**	0.097	0.197	0.124	0.229	0.106	0.209	0.149	0.198	0.124	0.228	0.097	0.195	0.108	0.215	0.143	0.259	0.122	0.228	0.168	0.267
BE	**0.374**	**0.241**	0.394	0.270	0.520	0.337	0.403	0.264	0.436	0.294	0.523	0.336	0.419	0.288	0.463	0.313	0.731	0.412	0.433	0.289	0.500	0.333
FR	**0.381**	**0.211**	0.439	0.233	0.507	0.290	0.411	0.220	0.440	0.216	0.510	0.290	0.431	0.234	0.429	0.260	0.855	0.384	0.466	0.242	0.519	0.295
DE	**0.440**	**0.418**	0.479	0.443	0.574	0.498	0.461	0.432	0.540	0.423	0.568	0.496	0.502	0.446	0.520	0.463	0.565	0.497	0.483	0.447	0.674	0.544
AVG	**0.304**	**0.265**	0.335	0.289	0.412	0.339	0.330	0.282	0.362	0.284	0.412	0.338	0.340	0.290	0.366	0.314	0.533	0.384	0.360	0.303	0.453	0.368

Transformer-based model

Linear-based model

CNN-based model

相比之下，使用注意力机制，基于transformer的模型更适合捕捉时间内和时间序列间数据之间的关系。

04 Long-term forecasting task

模型细节：将补丁长度设置为16，并将回顾序列的长度固定为96，预测长度变化{96，192，336，720}。较低的MSE或MAE表示更准确的预测结果。

Table 3. Full results of the long-term forecasting with exogenous variables. We compare extensive competitive models under different prediction lengths following the setting of iTransformer (Liu et al., 2023). The input sequence length is set to 96 for all baselines. Results are averaged from all prediction lengths. The character "." in the Transformers indicates the name of *former. "-" denotes out of memory (OOM) problem. The complete results are listed in the Appendix 7.

Model	TimeXer (Ours)		iTrans. (2023)		RLinear (2023)		PatchTST (2022)		Cross. (2022)		TiDE (2023)		TimesNet (2023a)		DLinear (2023)		SCINet (2022a)		Stationary (2022b)		Auto. (2021)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ECL	**0.336**	**0.414**	0.365	0.442	0.444	0.486	0.394	0.446	0.344	0.412	0.419	0.468	0.410	0.476	0.393	0.457	0.427	0.490	0.372	0.450	0.495	0.528
Weather	0.002	0.031	0.002	0.031	**0.002**	**0.029**	0.002	0.031	0.005	0.055	**0.002**	**0.029**	0.097	0.115	0.006	0.066	0.007	0.071	0.002	0.031	0.006	0.060
ETTh1	**0.074**	**0.210**	0.075	0.211	0.084	0.224	0.078	0.215	0.285	0.447	0.083	0.223	0.076	0.215	0.116	0.259	0.437	0.565	0.110	0.256	0.130	0.282
ETTh2	**0.183**	**0.337**	0.199	0.352	0.205	0.356	0.192	0.345	1.027	0.873	0.205	0.356	0.210	0.362	0.224	0.369	1.155	0.955	0.262	0.405	0.242	0.386
ETTm1	**0.051**	**0.169**	0.053	0.175	0.053	0.173	0.053	0.173	0.411	0.548	0.053	0.173	0.054	0.175	0.066	0.188	0.098	0.241	0.077	0.204	0.085	0.230
ETTm2	**0.116**	**0.252**	0.127	0.267	0.122	0.261	0.120	0.258	0.976	0.769	0.122	0.261	0.129	0.271	0.126	0.263	0.685	0.713	0.207	0.333	0.154	0.305
Traffic	**0.150**	**0.227**	0.161	0.246	0.324	0.412	0.173	0.253	-	-	0.324	0.411	0.171	0.264	0.323	0.404	0.447	0.500	0.361	0.361	0.302	0.353

长期多变量数据集包含大量的外生变量，并且并非所有的外生变量都与内生变量相关。TimeXer能够有效地捕捉有益的预测信息。

04 Abalation Studies

消融实验：进行了一系列消融实验来分析模型的不同组成部分对预测性能的影响，例如改变嵌入层的设计、调整补丁长度、以及分析交叉注意力和自注意力的作用。

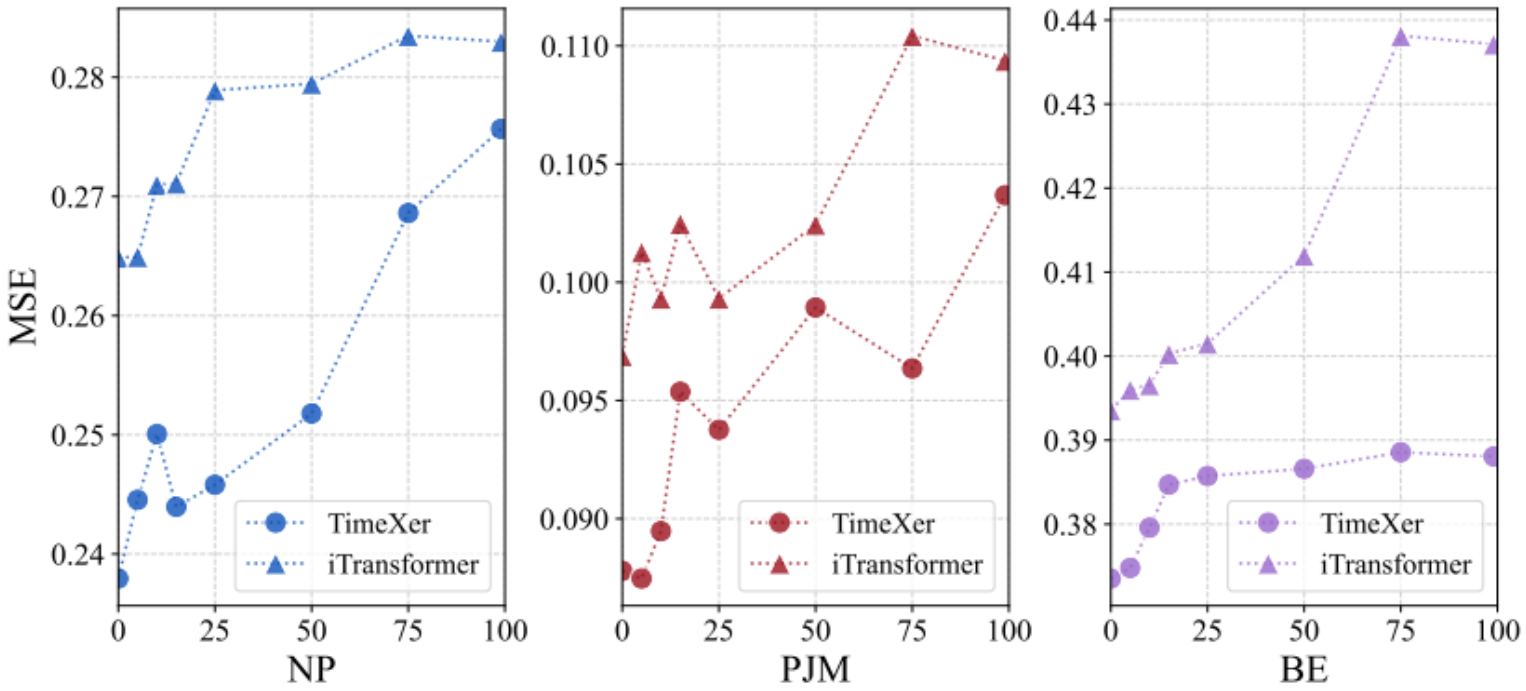


Figure 4. Forecasting performance with the masked exogenous series on three EPF datasets, simulating the missing data scenario.

使用随机掩码来探讨在输入低质量数据时的预测性能。使用完整的，高质量的历史数据作为内生变量，并通过从0%（即，对于外生变量使用完整的历史数据）到99%对外生变量进行掩码。

可以观察到，随着外生序列质量的降低，模型的能也随之降低。值得注意的是，当外源序列被少量掩盖时，TimeXer能够支持低质量的数据场景。

04 Varying Different Embeddings

Table 4. Ablation Results. *Ex.* and *En.* are abbreviations for Exogenous variable and Endogenous variable. *T* and *V* denote temporal token and variate token respectively. Results are averaged from all prediction lengths. Full results are listed in the Appendix.

	Design	En.	Ex.	ETTh2		ETTm2		Traffic	
				MSE	MAE	MSE	MAE	MSE	MAE
	Ours	T+V	V	**0.183**	**0.337**	**0.116**	**0.252**	**0.150**	**0.227**
针对外生变量	Replace	T+V	T	0.192	0.343	0.122	0.259	0.158	0.239
针对内生变量	**w/o**	w/o V	V	0.197	0.347	0.117	0.254	0.158	0.239
		w/o T	V	0.188	0.341	0.118	0.256	0.152	0.233

04 Analysis of Variate-wise Correlations —cross attention

将学习到的注意力图与具有最高和最低注意力得分的时间序列一起可视化

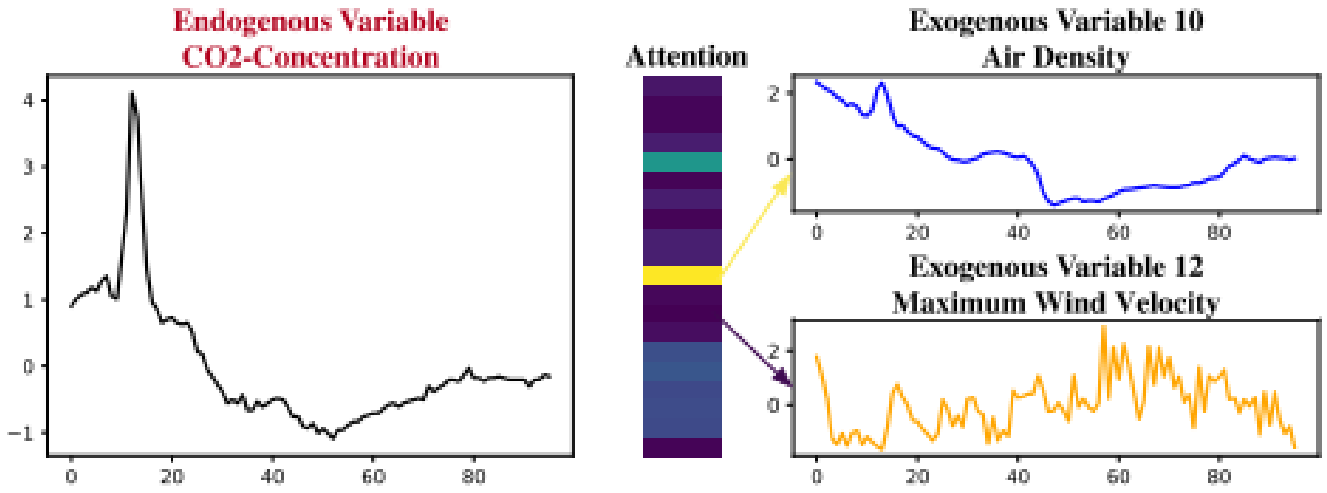


Figure 5. Visualization of learned attention map alongside the endogenous time series and the exogenous time series with highest and lowest attention scores.

对于内生变量CO2-浓度，其与空气密度之间确实存在很强的相关性，而最大风速的影响相对较小，颜色越深关系越小

TimeXer能够区分与内生变量表现出相似性的外生变量和几乎没有相关性的外生变量。前者注意力得分更大

04 Representation Analysis表示分析

CKA相似度：比较两个特征空间的相似性

$$\mathrm{CKA} = \frac{\mathrm{Tr}(\mathbf{K}_X \mathbf{K}_Y^T)}{\sqrt{\mathrm{Tr}(\mathbf{K}_X \mathbf{K}_X^T)\mathrm{Tr}(\mathbf{K}_Y \mathbf{K}_Y^T)}}$$

CKA相似度的值范围在-1到1之间，值越接近1表示两组特征表示越相似，值越接近-1表示越不相似。

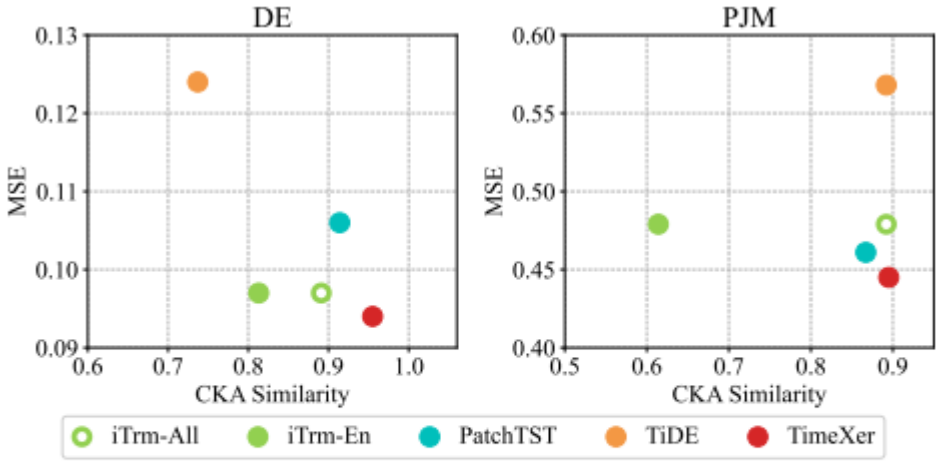


Figure 7. Comparison of representation CKA similarity between TimeXer and competitive baselines. *iTrm-All* denotes the series representation of all variables learned by iTransformer, *iTrm-En* is the learned series representation of the endogenous variable.

04 Increasing Exogenous Look-back Length

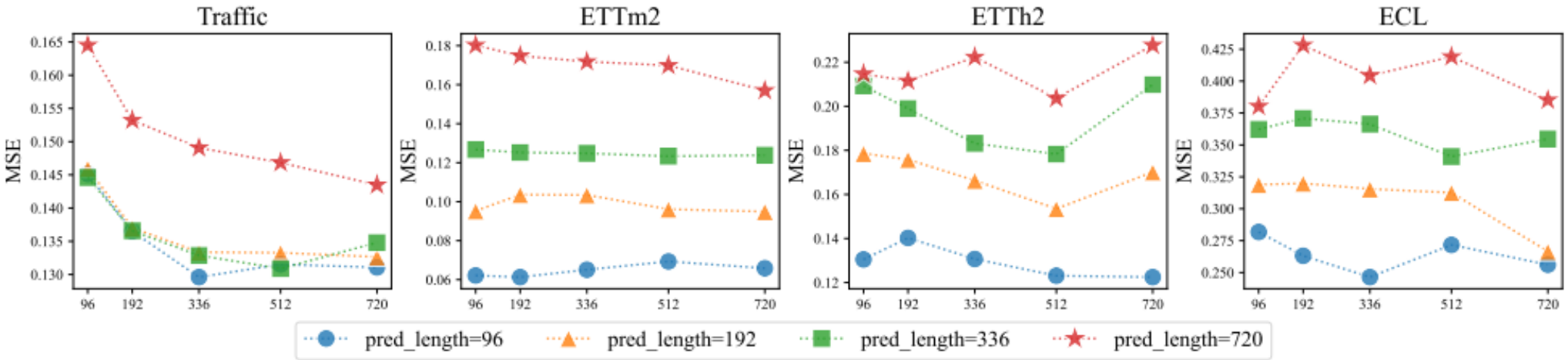


Figure 6. Forecasting performance with the look-back length of exogenous variables varying from $\{96, 192, 336, 512, 720\}$ with a fixed look-back length of the endogenous variable to 96. Different styles of lines represent different prediction lengths. In most cases, the forecasting performance benefits from the increase of exogenous look-back length.

将内生变量的长度固定为96，并增加外生变量的长度。结果表明，延长外生变量的回顾长度确实可以提高预测性能。但是到512时，都会出现一定的波动

04 扩展到多变量预测

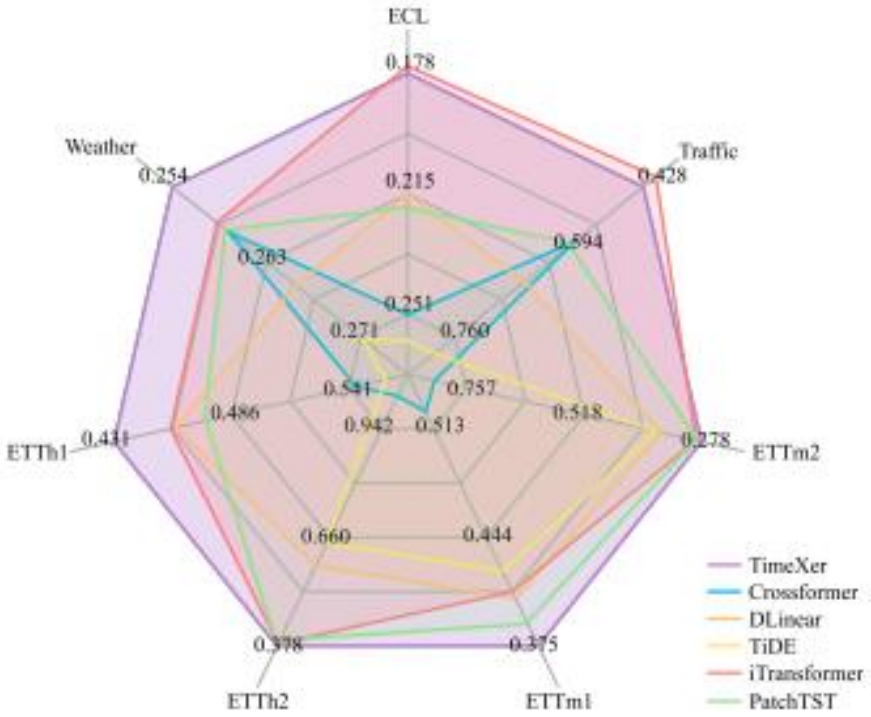


Figure 8. Performance Comparison between TimeXer with state-of-art models. Full results are listed in Appendix 8.

原则上，多元预测可以看作是对多元数据中的每个变量的独立预测，对于每个变量，其他变量是外生的，以便于预测

05 Conclusion

TimeXer模型在时间序列预测方面具有创新性和有效性，但也面临一些局限性，主要体现在效率、模型复杂度、数据适应性、多模态处理和长期依赖捕捉等方面，这些局限性制约了模型在复杂实际场景中的应用和性能提升，需要在未来研究中加以改进。具体如下：

1. 效率方面

- 在处理少量外生变量时，效率优势不明显，相较于部分基线模型，**计算资源利用效率有待提高**，可能造成不必要的计算成本和时间消耗。

2. 模型复杂度与可解释性方面

- 模型结构复杂，多种嵌入和注意力机制的运用使其内部工作机制难以理解，可解释性差，在对透明度要求高的场景中应用受限。
- **超参数众多，调参过程复杂且耗时**，不当的超参数设置会影响性能，对资源有限的用户不友好。

3. 数据假设与实际数据差异方面

- 对数据分布假设敏感，实际数据分布变化时性能可能受影响，如在异常值多或季节性变化剧烈的数据集中表现不佳。
- 处理极端值和异常值能力未充分研究，在实际数据含此类值时，可能导致预测偏差，影响稳定性和可靠性。

4. 多模态数据处理能力方面

主要针对内生单变量情况，对内生变量多模态分布的时间序列处理能力不足，无法准确捕捉复杂变化，影响预测性能。

5. 长期依赖捕捉方面

尽管有优势，但处理极长序列时仍可能受信息衰减或梯度消失影响，难以准确建模长期依赖关系，影响长期预测准确性。